INDEX

ACTIVITY - 1
LIMIT AND CONTINUITY

OBJECTIVE:

To find analytically the limit of a function $f(x)$ at $x=0$ and also to check the continunity of the function at that point.

Consider the function

$$f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \text{ and} \\ 6, & x = 2 \end{cases}$$

Check its continuity at $x=2$

PREVIOUS KNOWLEDGE:

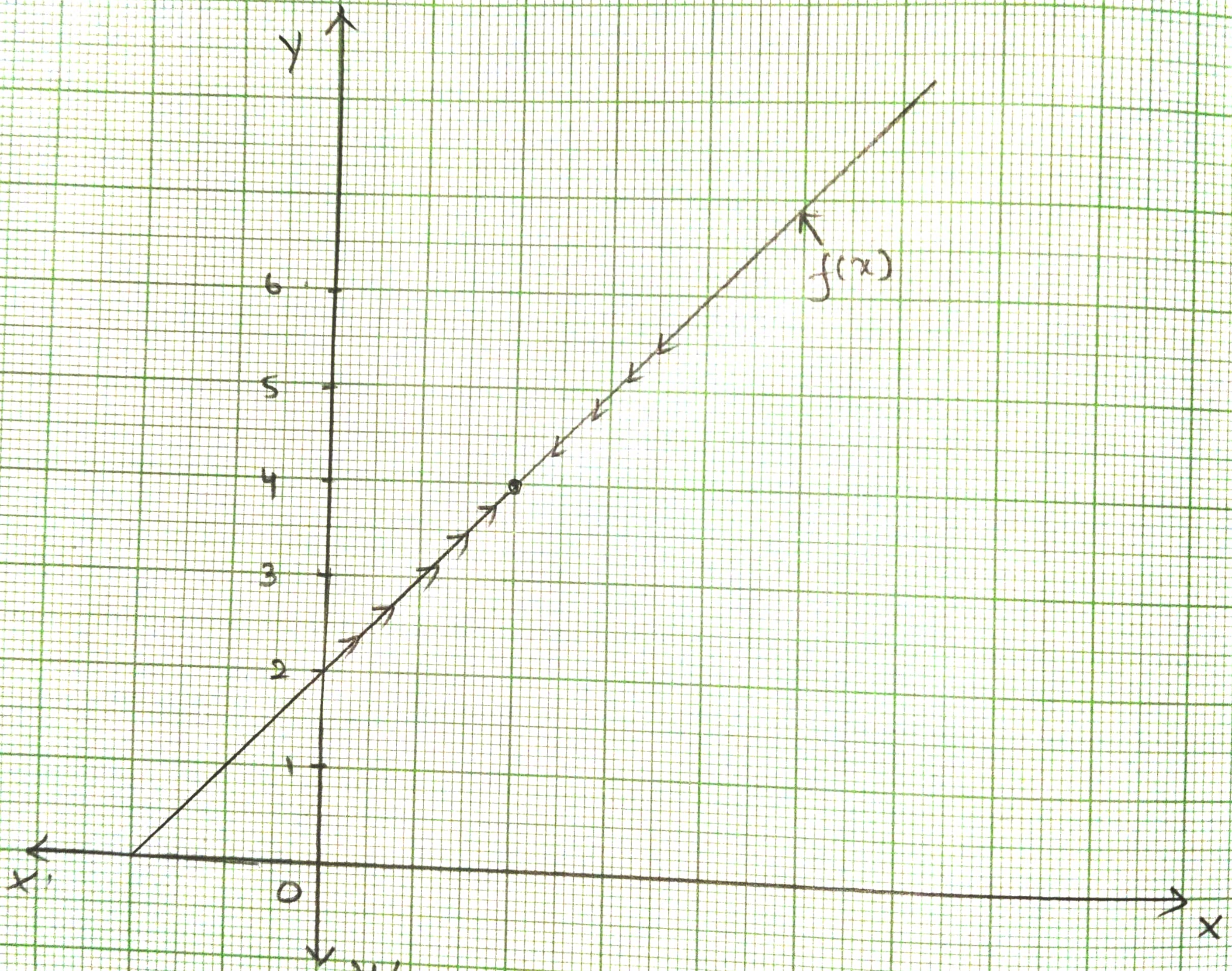
A function $f(x)$ is said to be continuous at a point a in its domain if $\lim_{x \to a} f(x) = f(a)$

MATERIALS REQUIRED:

- Graph Paper
- Geometry Box
- Adhesive
- Pair of scissors.

PROCEDURE:

(i) Given function is $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ 6, & x = 2 \end{cases}$



y
6
5
4
3
2
1
0
f(x)
X'
X
Y'

ii) We have to check the continuity of $f(x)$ at $x=2$.
iii) $f(x)$ has continuity at $x=2$ if $\lim_{x \to 2} f(x) = f(2)$
iv) As x approaches 2, the variable x takes values which are less than or greather than 2 and are very close to 2.
v) when $x \neq 2$, $f(x) = \frac{x^2-4}{x-2} = \frac{(x-2)(x+2)}{x-2} = x+2$

vi) The following table shows the values of $f(x)$ as $x \to 2$ and $x < 2$.

$x \to 2$ and $x < 2$	1.9	1.92	1.995	1.9993	1.99998...
$f(x)$	3.9	3.92	3.995	3.9995	3.9998...

Thus $x \to 2$ from left, the function $f(x)$ approaches 4
vii) The following table shows the values of $f(x)$ as $x \to 2$ and $x > 2$

$x \to 2$ and $x > 2$	2.1	2.03	2.005	2.0009	2.00007...
$f(x)$	4.1	4.03	4.005	4.0009	4.00007...

Thus $x \to 2$ from eight, the function $f(x)$ approaches 4
viii) Take a rectangular piece of graph paper.
ix) Draw the graph of $f(x)$ on the graph paper.

OBSERVATIONS:
i) From the above tables we have
$\lim_{x \to 2^-} f(x) = 4$ and $\lim_{x \to 2^+} f(x) = 4$

ii) $\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x) = 4$

$\therefore \lim_{x \to 2^-} f(x)$ exists and is equal to 4

iii) Also $f(x) = 6$

iv) $\lim_{x \to 2} f(x) \neq f(2)$

$\therefore f(x)$ is not continuous at $x = 2$.

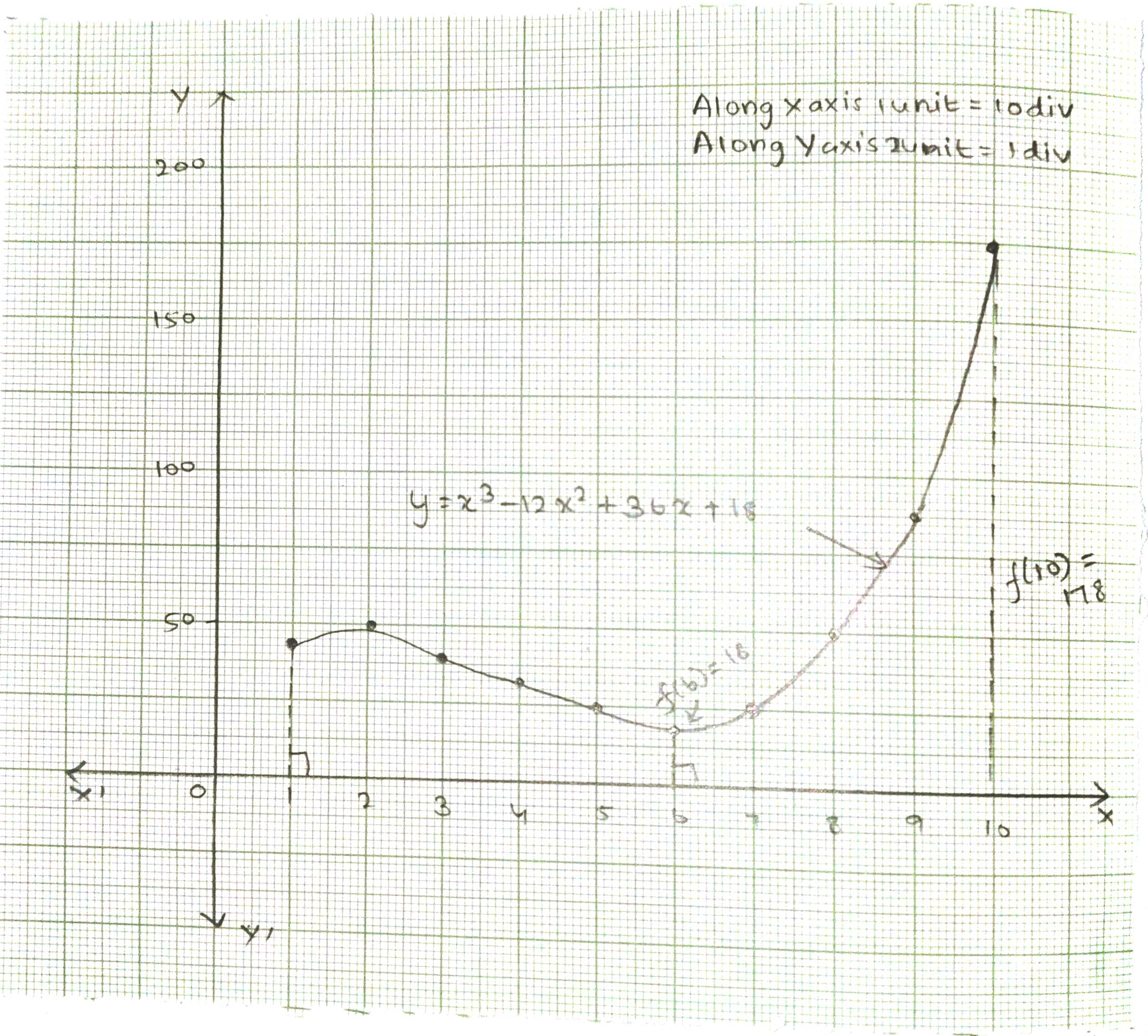
v) The curves of $f(x)$ is having a break at $x = 2$

RESULT:

For given function $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ 6, & x = 2 \end{cases}$, we have

$\lim_{x \to 2} f(x) = 4$. Also, the function $f(x)$ is not continuous

at $x = 2$.


Along x axis 1unit = 10div
Along Y axis 2unit = 1div
Y
200
150
100
50
X'
0
1
2
3
4
5
6
7
8
9
10
X
Y'
y = x^3 - 12x^2 + 36x + 18
f(6) = 18
f(10) = 178

ACTIVITY - 2

ABSOLUTE MAXIMUM AND MINIMUM VALUES

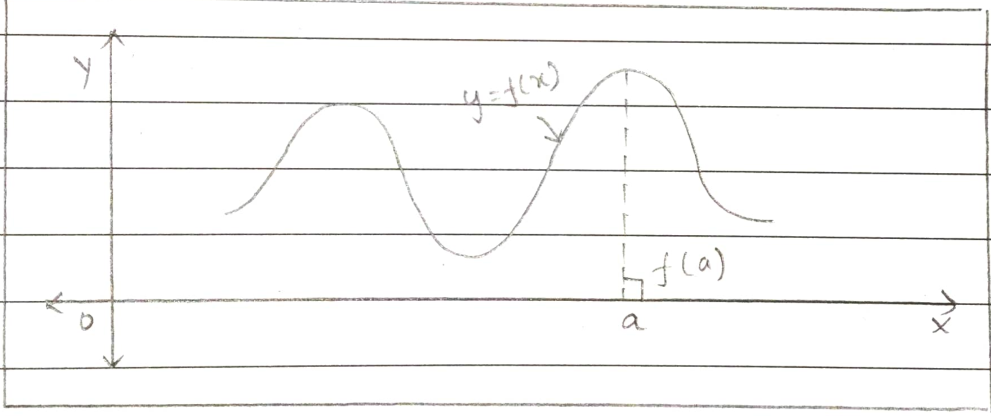
OBJECTIVE:

To understand the concept of absolute maximum and minimum values of a function in a given closed interval through its graph. consider the graph:

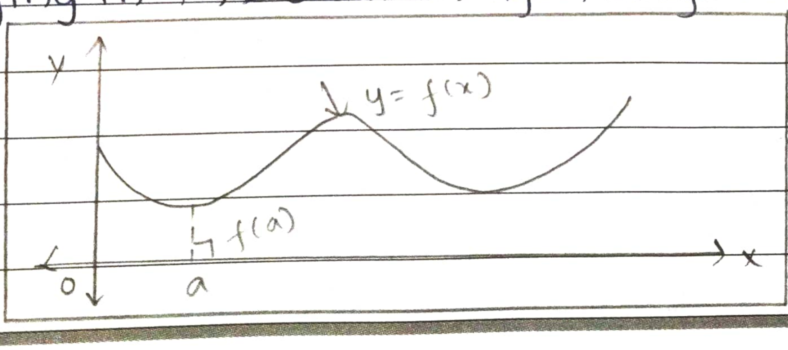
$$f(x) = x^3 - 12x^2 + 36x + 18, \ x \in [1,10].$$

PREVIOUS KNOWLEDGE:

i) A function $f(x)$ is said to have absolute maximum value at $x=a$ if $f(a)$ is the greatest of all its values of x lying in the domain of the function.



ii) A function $f(x)$ is said to have absolute minimum value at $x=a$ if $f(a)$ is the smallest of all its values for values of x lying in the domain of the function.



iii) If $x_1, x_2, \dots x_n$ are the values of x where $f'(x)=0$ or where $f(x)$ is not derivable then the greatest and least values of in the set $\{f(a), f(x_1), f(x_2), \dots f(x_n), f(b)\}$ are respectively the absolute maximum value and absolute minimum value of $f(x)$ on the closed interval $[a,b]$

MATERIALS REQUIRED:

- Graph paper
- Geometry paper
- Pair of scissors
- Adhesive.

PROCEDURE:

i) Given function is $f(x) = x^3 - 12x^2 + 36x + 18,\ x \in [1,10]$

ii) $f'(x) = 3x^2 - 24x + 36$

iii) $f'(x) = 0 \Rightarrow 3x^2 - 24x + 36 = 0 \Rightarrow x^2 - 8x + 12 = 0 \Rightarrow x = 2, 6$

iv) we find the values of $f(x)$ at $x = 1, 2, 6$ and 10

$f(1) = 1^3 - 12(1)^2 + 36(1) + 18 = 1 - 12 + 36 + 18 = 43$

$f(2) = 2^3 - 12 \times 4 + 36 \times 2 + 18 = 8 - 48 + 72 + 18 = 50$

$f(6) = 6^3 - 12 \times 6^2 + 36 \times 6 + 18 = 216 - 432 + 216 + 18 = 18$

$f(10) = 10^3 - 12 \times 10^2 + 36 \times 10 + 18 = 1000 - 1200 + 360 + 18 = 178$

v)

x	1	2	6	10
$f(x)$	43	50	18 - min	178 - max

Absolute max value = 178 at $x = 10$ and absolute min value 18 at $x = 6$

vi) Now, we shall verify these values using the graph of $f(x) = x^3 - 12x^2 + 36x + 18,\ x \in [1,10]$

vii) Take a rectangular piece of graph paper and draw the X-axis and Y-axis.

viii) We find some points on the graph of $f(x) = x^3 - 12x^2 + 36x + 18$, $x \in [1,10]$

x	$y = f(x) = x^2 - 12x^2 + 36x + 18$
1	$1^3 - 12(1)^2 + 36 + 18 = 1 - 12 + 36 + 18 = 43$
2	$2^3 - 12 \times 2^2 + 36 \times 2 + 18 = 8 - 48 + 72 + 18 = 50$
3	$3^3 - 12 \times 3^2 + 36 \times 3 + 18 = 27 - 108 + 108 + 18 = 45$
4	$4^3 - 12 \times 4^2 + 36 \times 4 + 18 = 64 - 192 + 144 + 18 = 34$
5	$5^3 - 12 \times 5^2 + 36 \times 5 + 18 = 125 - 300 + 180 + 18 = 28$
6	$6^3 - 12 \times 6^2 + 36 \times 6 + 18 = 216 - 432 + 216 + 18 = 18$
7	$7^3 - 12 \times 7^2 + 36 \times 7 + 18 = 343 - 588 + 252 + 18 = 25$
8	$8^3 - 12 \times 8^2 + 36 \times 8 + 18 = 512 - 768 + 288 + 18 = 50$
9	$9^3 - 12 \times 9^2 + 36 \times 9 + 18 = 729 - 972 + 324 + 18 = 99$
10	$10^3 - 12 \times 10^2 + 36 \times 10 + 18 = 1000 - 1200 + 360 + 18 = 178$

ix) The points (1,43), (2,50), (3,45), (4,34), (5,23), (6,18), (7,25), (8,50), (9,99), (10,178) are on the graph of $f(x) = x^3 - 12x^2 + 36x + 18$, $x \in [1,10]$.

(x) Plot these points on the graph paper and join these points by a free hand curve. This is the graph of $f(x) = x^3 - 12x^2 + 36x$ $x \in [1,10]$.

OBSERVATIONS:

From the graph of $y = x^3 - 12x^2 + 36x + 18$, $x \in [1,10]$ we observe that the absolute maximum value is 178 at $x = 10$ and absdute minimum value is 18 at $x = 6$.

RESULT:

For the given function $f(x) = x^3 - 12x^2 + 36x + 18$, $x \in [1,10]$ absolute maximum value $= 178$ at $x = 10$ and absolute minimum value $= 18$ at $x = 6$.


PERIMETER = 36 units

ACTIVITY – 3

APPLICATION OF MAXIMA AND MINIMA

OBJECTIVE :

To verify that amongst all the rectangles of same perimeters, the square has maximum area. Consider a rectangle of perimeter 36 units.

PREVIOUS KNOWLEDGE :

i) The function $f(x)$ has maximum value at $x=a$, if $f'(a)=0$ and $f''(a)<0$.

ii) The function $f(x)$ has minimum value at $x=a$, if $f'(a)=0$ and $f''(a)>0$.

MATERIALS REQUIRED

- Geometry Box.
- Graph Paper
- Adhesive.

PROCEDURE :

i) PERIMETER of the rectangle is 36 units.

ii) Draw a rectangle ABCD

iii) Let side $AB = x$ units and side $BC = y$ units

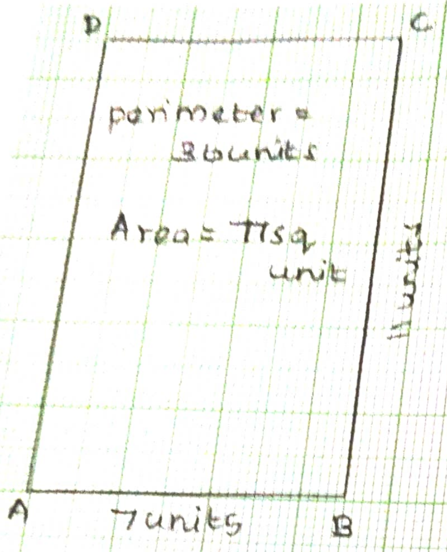
iv) Perimeter $= 2(x+y)$ units and side $BC = y$ units $\therefore 2(x+y)=36$

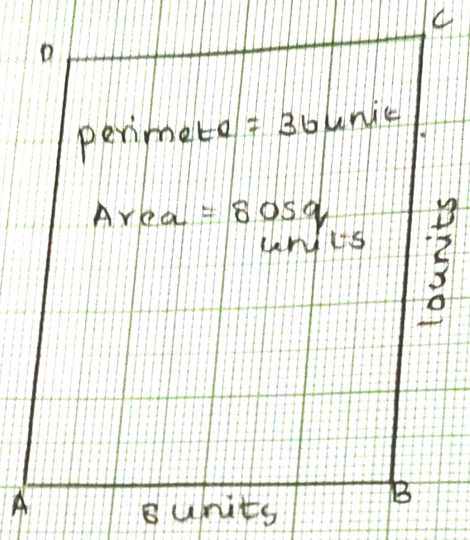
$x+y=18$, $y=18-x$

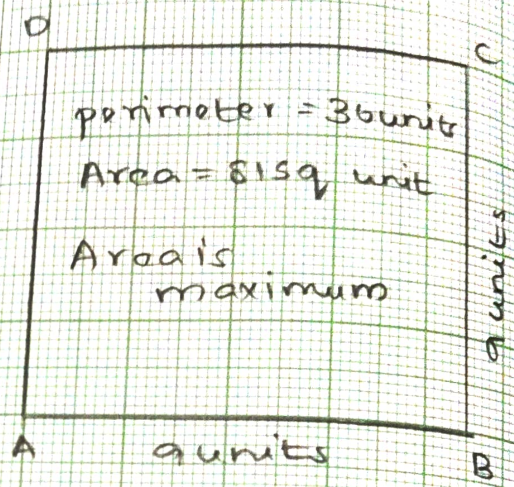
v) Let A sq. units be the area of the rectangle ABCD

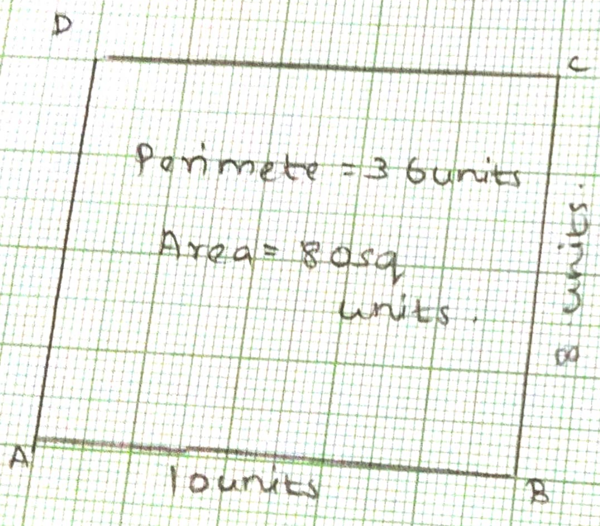
$\therefore A = xy = x(18-x) = 18x - x^2$

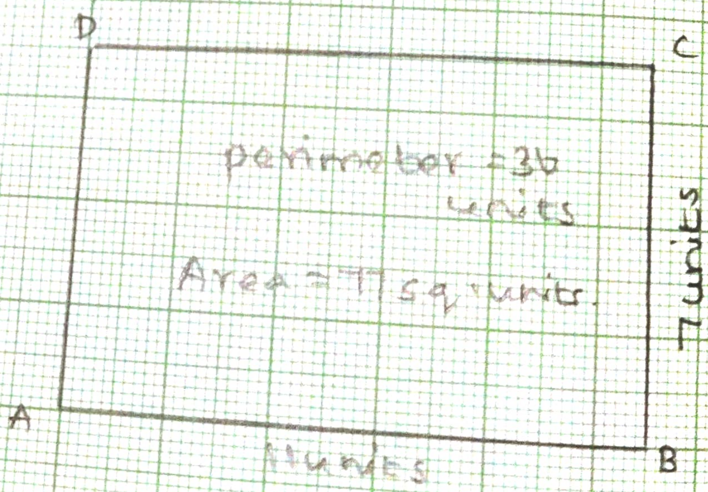
$\therefore A = 18x - x^2$


D
C
perimeter = 36units
Area = 77sq unit
11 units
A
7units
B


D
C
perimete = 36unit
Area = 80sq units
10units
A
8 units
B


D
C
perimeter = 36unit
Area = 81sq unit
Area is maximum
9 units
A
9units
B


D
C
Perimete = 36units
Area = 80sq units.
8 units
A
10units
B


D
C
perimeter = 36 units
Area = 77sq units.
7units
A
11units
B

vi) we have, $A = 18x - x^2$

we have to find the value of x for which the value of area A of the rectangle is maximum.

$$\frac{dA}{dx} = 18 - 2x$$

$$\frac{d^2A}{dx^2} = -2$$

$$\frac{dA}{dx} = 0 \Rightarrow 18 - 2x = 0 \Rightarrow x = 18/2 \Rightarrow x = 9$$

$$\left.\frac{d^2A}{dx^2}\right|_{x=9} = -2 < 0$$

$\therefore$ A is maximum when $x = 9$

Also, $x = 9 \Rightarrow y = 18 - 9 = 9$

vii) The area of the rectangle is maximum when side $AB = x$ units = 9 units and side $BC = y$ units = 9 units.

Rectangle is a square of side 9 units.

Now, we shall verify this by taking different values of x.

viii) Table 1

Sl No.	x	$y = 18 - x$	$A = xy$
1	7	$y = 18 - 4 = 11$	$A = xy = 7 \times 11 = 77$
2	8	$y = 18 - 8 = 10$	$A = 8 \times 10 = 80$
3	9	$y = 18 - 9 = 9$	$A = 9 \times 9 = 81$
4	10	$y = 18 - 10 = 8$	$A = 10 \times 8 = 80$
5	11	$y = 18 - 11 = 7$	$A = 11 \times 7 = 77$

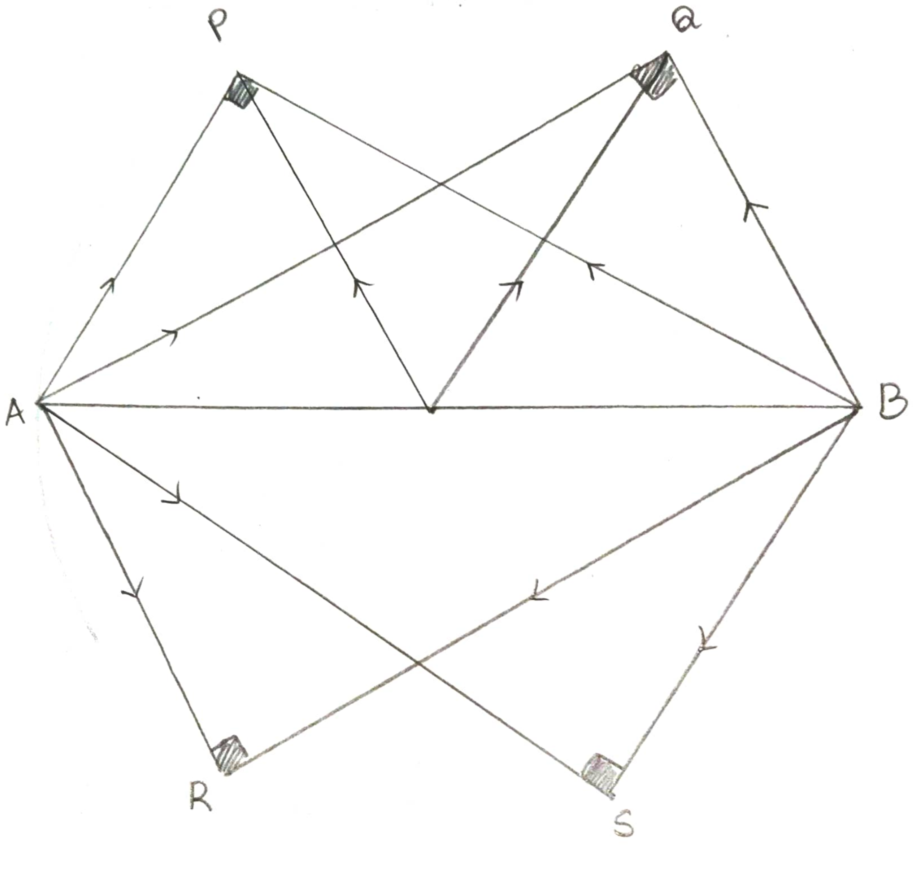
OBSERVATIONS:

i) Table 1 shows the values of A for values of x which are less than 9 and more than 9.

ii) we observe that the values $A = 81$ is maximum for $x = 9$ and for other values of x, the value of A is less than 81.

RESULT:

The area of rectangle of perimeter 36 units is maximum when the rectangle is a square of side 9 units.


P
Q
A
B
R
S

ACTIVITY 4

OBJECTIVE

To verify that angle in a semicircle is a right angle using vector method.

MATERIALS REQUIRED:

Chart Paper, white paper, Adhesive, pens, geometry box, eraser.

METHOD OF CONSTRUCTION:

1. Take a chart paper.
2. On the chart paper, paste a white paper of the same size using an adhesive.
3. On this paper, draw a circle with centre O and radius 4cm.
4. Mark points O, A, B, P and Q. Join OP, OA, OB, AP, AQ BQ, OQ and BP.
5. Represent OA, OB, OP, AP, BP, OQ, AQ and BQ as vectors

DEMONSTRATION:

1. Using a protractor, measure the angle between the vectors $\overrightarrow{AP}$ and $\overrightarrow{BP}$ i.e $\angle APB = 90°$
2. Similarly, the angle between the vectors $\overrightarrow{AQ}$ and $\overrightarrow{BQ}$ i.e $\angle AQB = 90°$.
3. Repeat the above process by taking some more points R, S, T... on semicircles forming vectors $\overrightarrow{AR}$, $\overrightarrow{BR}$, $\overrightarrow{AS}$, $\overrightarrow{BS}$, $\overrightarrow{AT}$, $\overrightarrow{BT}$.... etc. i.e any angle formed between vector in semicircle is 90°.

OBSERVATION:

By actual movement

$|\vec{OP}| = |\vec{OA}| = |\vec{OB}| = |\vec{OQ}| = r = a = p = 4$

$|\vec{AP}| = |\vec{p} + \vec{a}|$, $|\vec{BP}| = |\vec{p} - \vec{a}|$, $|\vec{AB}| = |2\vec{a}|$

$|\vec{AQ}| = |\vec{r} + \vec{a}|$, $|\vec{BQ}| = |\vec{r} - \vec{a}|$

$|\vec{AP}|^2 + |\vec{BP}|^2 = 4|\vec{a}|^2 = |\vec{AB}|^2$

$|\vec{AQ}|^2 + |\vec{BQ}|^2 = 4|\vec{a}|^2 = |\vec{AB}|^2$

So, $\angle APB = 90°$ and $\vec{AP} \cdot \vec{BP} = 0$

$\angle AQB = 90°$ and $\vec{AQ} \cdot \vec{BP} = 0$

Similarly for points R, S, T, $|\vec{AB}|^2 = |\vec{BR}|^2 = |AB|^2$ and so on.

$\angle ARB = 90°$, $\angle ASB = 90°$, $\angle ATB = 90°$

RESULT:

Angle in a semicircle is a right angle.

ACTIVITY - 5

OBJECTIVE:

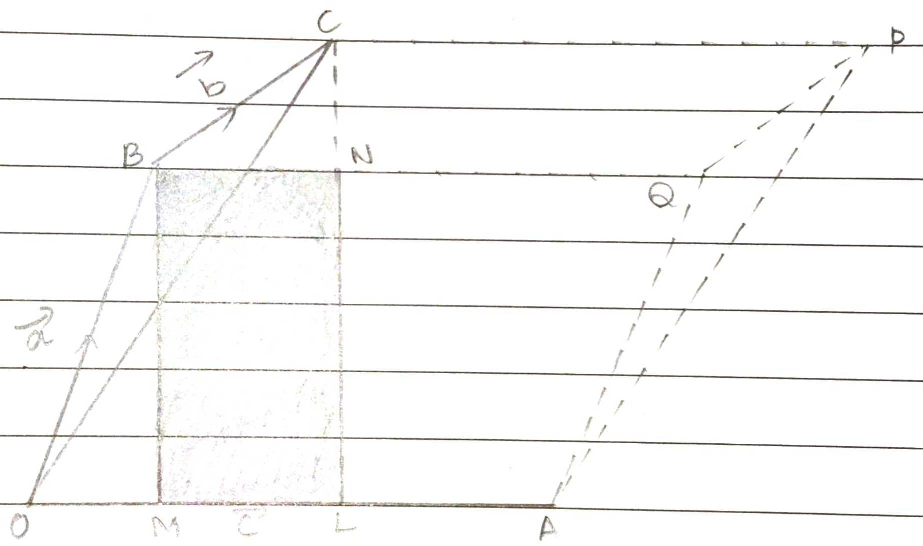
To verify geometrically that $\vec{c} \times (\vec{a}+\vec{b}) = |\vec{c} \times \vec{a}| + |\vec{a} \times \vec{b}|$

MATERIALS RECQIRED:

Geometry Box, white paper, sketch pen.

METHOD OF CONSTRUCTION:

1. Paste an A_4 size sheet.
2. Draw a line segment OA = (6cm say) and let it represent $\vec{c}$.
3. Draw another line segment OB (= 4cm say) at angle (60°) with OA. Let $\vec{OB} = \vec{a}$



4. Draw BC (= 3cm say) making an angle (say 30°) with $\vec{OA}$. Let $\vec{BC} = \vec{b}$
5. Draw perpendicular BM, CL and BN
6. Complete parallelogram OAPC, OAQB and BQPC

DEMONSTRACTION:

1. $\vec{OC} = \vec{OB} + \vec{BC} = \vec{a} + \vec{b}$ and let angle $\angle COA = \alpha$
2. $|\vec{c}(\vec{a} \times \vec{b})| = |\vec{c}||\vec{a} + \vec{b}| \sin\alpha$ = area of praralellogram OAPC.
3. $|\vec{c} \times \vec{a}|$ = areaof parallelogram OAQB.
4. $|\vec{c} \times \vec{b}|$ = area of parallelogram BQPC.
5. Areaof parallelogram OAPC = (OA)(CL)

$= (OA)(LN + NC) = (OA)(BM + NC)$

$= (OA)(BM) + (OA)(NC)$

= Area of parallelogram OAQB + Areaof parallelogram BQPC

$= |\vec{c} \times \vec{A}| + |\vec{c} \times \vec{b}|$

So $|\vec{c} \times (\vec{a} + \vec{b})| - |\vec{c} \times \vec{a}| + |\vec{c} \times \vec{b}|$

Direction of each of these vectors $\vec{c} \times (\vec{a} + \vec{b})$ is $\perp$ to the same plane.

So $\vec{c} \times (\vec{a} + \vec{b}) = \vec{c} \times \vec{a} + \vec{c} \times \vec{b}$

OBSERVATION:

$|\vec{c}| = |\vec{OA}| = OA = 6cm$

$|\vec{a} + \vec{b}| = |\vec{OC}| = OC = 6.7cm$

$CL = 5cm$

$|\vec{c} \times (\vec{a} + \vec{b})|$ = Areaof parallelogram OAPC

$= (OA)(CL) = 30$ sq units (i)

$|\vec{c} \times \vec{a}|$ = Areaof parallelogram OAQB

$= (OA)(BM) = 21\ cm^2$ (ii)

$|\vec{c} \times \vec{b}|$ = Area of parallelogram BQPC

$= (OA)(CN) = 6cm \times 1.5cm = 9cm^2$ (iii)

From (i)(ii) and (iii)

Areaof parallelogram OAPC = Areaof parallelogram OAQB

+

Areaof parallelogram BQPC

Thus,

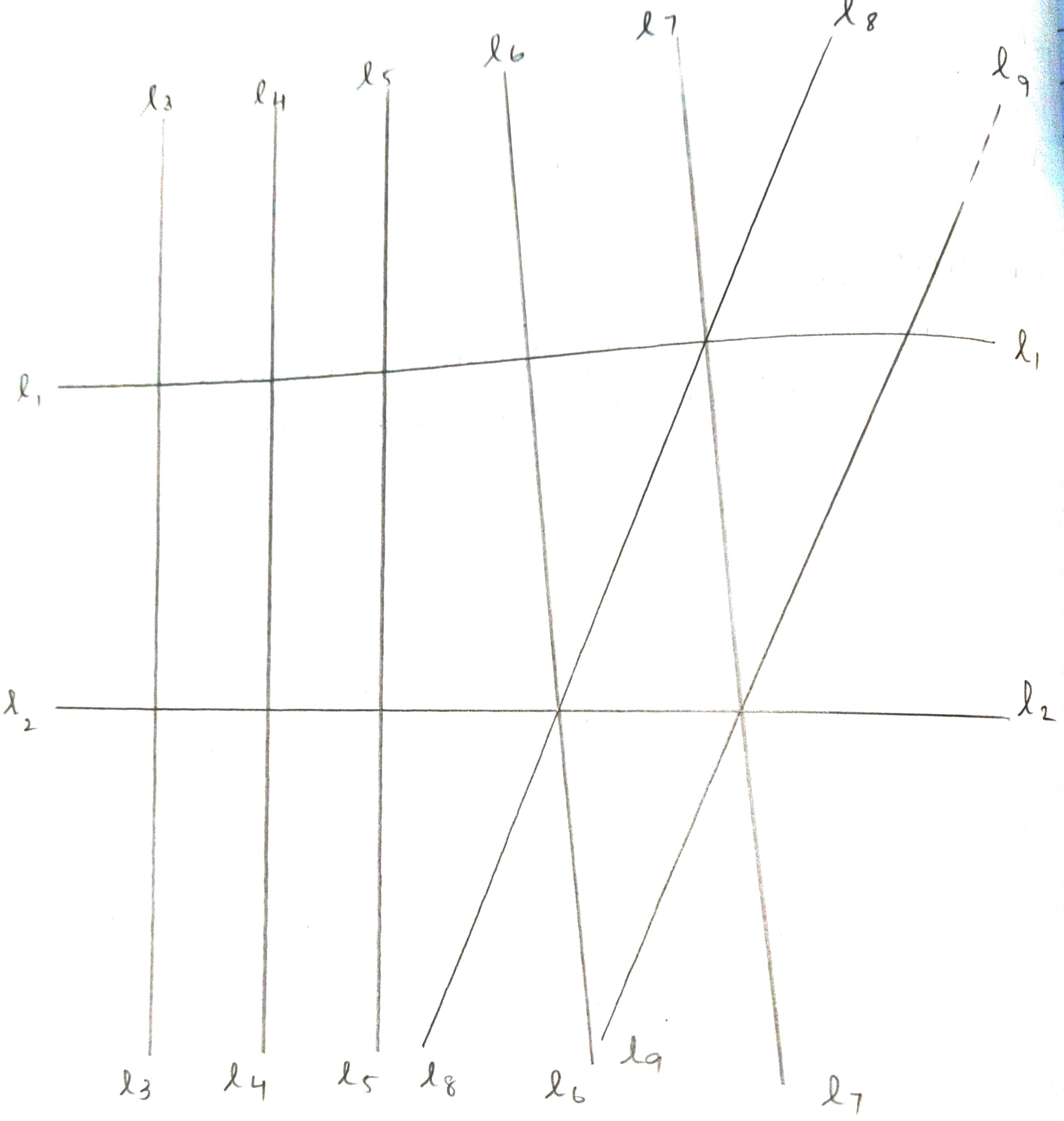
$|\vec{c} \times (\vec{a}+\vec{b})| = |\vec{c} \times \vec{a}| + |\vec{c} \times \vec{b}|$

$|\vec{c} \times \vec{a}|$, $|\vec{c} \times \vec{b}|$ and $\vec{c} \times (\vec{a}+\vec{b})$ are all in the direction perpendicular to the plane of paper.

Therefore $\vec{c} \times (\vec{a}+\vec{b}) = \vec{c} \times \vec{a} + \vec{c} \times \vec{b}$

APPLICATION:

Through the activity, distributive property of vector multiplication over addition can be explained.



l_3
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l_5
l_6
l_7
l_8
l_9
l_1
l_1
l_2
l_2
l_3
l_4
l_5
l_8
l_6
l_9
l_7

ACTIVITY - 6

NON - EQUIVALENCE RELATION

OBJECTIVE

To verify that the relation R on the set L of all lines in a plane defined by $R = \{(l, m) : l \perp m\}$ is not an equivalence relation consider nine lines in the plane.

PREVIOUS KNOWLEDGE:

1) Any subset of $A \times A$ is called a relation on the set A, where A is a non empty set.
2) A relation R on a non-empty set A is called a reflexive relation if $(a, a) \in R$ for all $a \in A$.
3) A relation R on a non empty set A is called a symmetric relation if $(a, b) \in R$, then $(b, a) \in R$.
4) A relation R on a non empty set A is called a transitive relation if $(a, b), (b, c) \in R$ then $(a, c) \in R$.
5) A relation R on a non empty set A is called an equivalence relation if it is reflexive, symmetric and transitive.

MATERIALS REQUIRED :

- Geometry Box
- Coloured paper
- Pair of scissors.

PROCEDURE:

i) we have to consider nine lines.

ii) Draw 9 lines on the paper so that some of them are parallel, some are perpendicular and some are neither parallel nor perpendicular.

OBSERVATION:

i) set L has lines l_1 to l_9

ii) l_1 and l_2 are parallel lines.

iii) l_3, l_4 and l_5 are parallel lines.

iv) l_1 is perpendicular to l_3, l_4 and l_5

v) l_2 is perpendicular to l_3, l_4 and l_5

vi) No lines can be perpendicular to itself $\therefore (l_1, l_i) \notin R$ for $i = 1, 2 \ldots$

vii) If a line l is perpendicular to line m then m is perpendicular to l.

viii) $\therefore R = \{(l_1, l_3)(l_1, l_4)(l_1, l_5)(l_3, l_1)(l_4, l_1)(l_5, l_1)(l_2, l_3)$
$(l_2, l_4)(l_2, l_5)(l_5, l_2)(l_4, l_2)(l_3, l_2)\}$

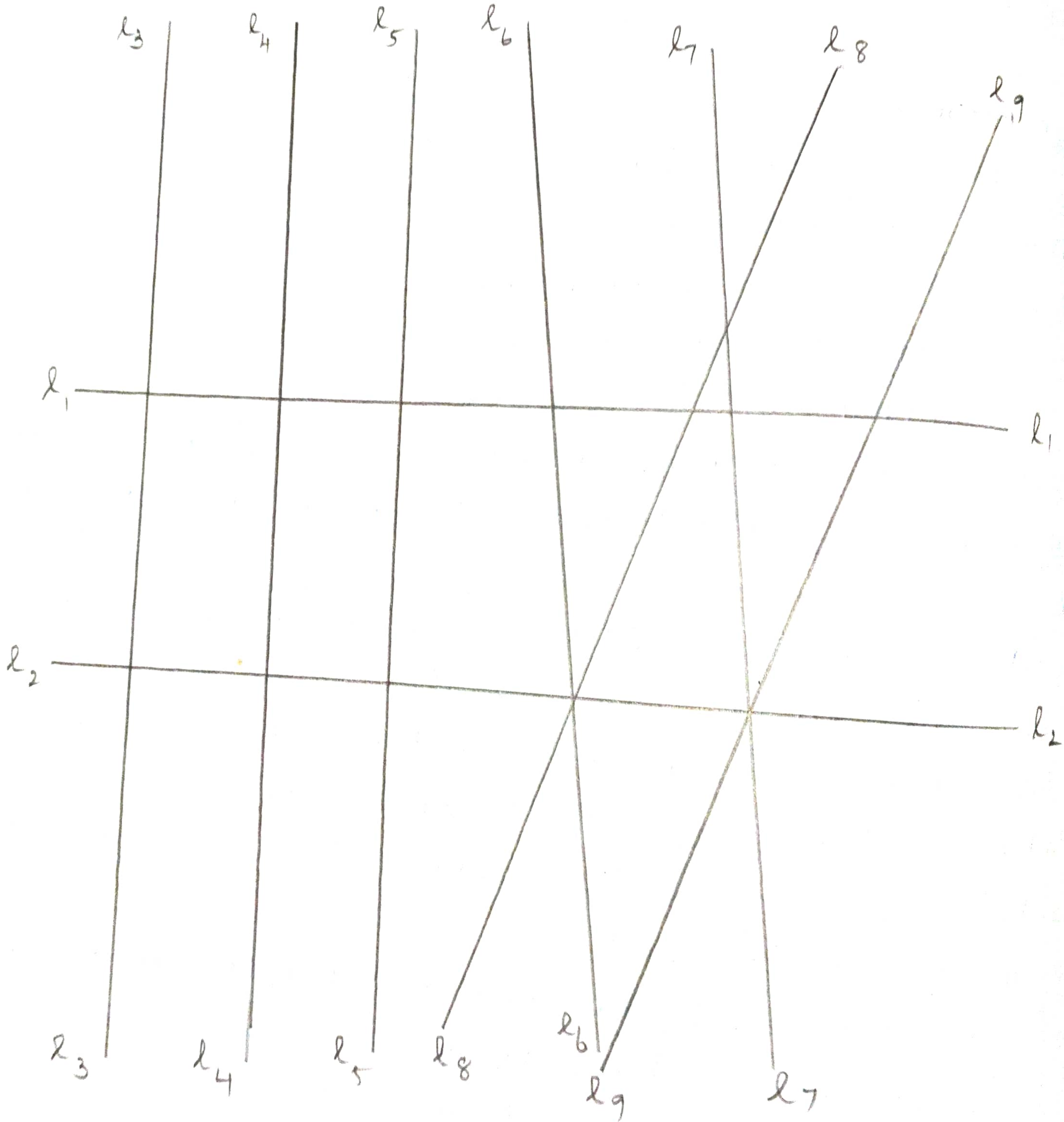
ix) Since $(l_1, l_2) \notin R$, the relation R is not reflexive.

x) We have $(l_1, l_3)(l_3, l_1) \in R$, $(l_1, l_4)(l_4, l_1) \in R$
$(l_2, l_5)(l_5, l_2) \in R$ $(l_1, l_5)(l_5, l_1) \in R$ $(l_2, l_3)(l_3, l_2) \in R$,
$(l_2, l_4)(l_4, l_2) \in R$. $\therefore$ R is symmetric.

xi) $l_1 \perp l_3$ and $l_3 \perp l_2$ but l_1 is not perpendicular to l_2
$\therefore (l_1, l_3)(l_3, l_2) \in R$ but $(l_1, l_2) \notin R$
$\therefore$ the relation R is not transitive.

RESULT:

Since R is neither reflective nor transitive, it is not an equivalence relation.



l_3
l_4
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l_6
l_7
l_8
l_9
l_1
l_1
l_2
l_2
l_3
l_4
l_5
l_8
l_6
l_9
l_7

ACTIVITY - 7

EQUIVALENCE RELATION

OBJECTIVE:

To verify that the relation R on the set L of all lines in a plane defined by $R = \{(l, m) \ l$ and m are either parallel or co-incident$\}$ is an equivalence relation. Consider nine lines in the plane.

PREVIOUS KNOWLEDGE:

i) Any subset of $A \times A$ is called a relation on the set A, where A is a non empty set.

ii) A relation R on a non empty set A is called a reflexive relation if $(a, a) \in R$ for all $a \in A$.

iii) A relation R on a non empty set A is called a symmetric relation if $(a, b) \in R$ then $(b, a) \in R$.

iv) A relation R on a non empty set A is called a transitive relation if $(a, b) \in R$, $(b, c) \in R$ then $(a, c) \in R$.

v) A relation R on a non empty set A is called an equivalence relation if it is reflective, symmetric and transitive.

MATERIALS REQUIRED:

Geometric box.

Pencils.

PROCEDURE:

i) We have to consider nine lines.

ii) Draw 9 lines on the paper so that some of them are parallel, some are perpendicular and some are neither parallel nor perpendiular.

OBSERVATIONS:

i) Set L has lines l_1 to l_9.

ii) l_1 and l_2 are parallel lines.

iii) l_3, l_4 and l_5 are parallel lines.

iv) l_1 is perpendicular to l_3, l_4 and l_5.

v) l_2 is perpendicular to l_3, l_4 and l_5.

vi) A line always coincide with itself : $(l_i, l_i) \in R$ for $i = 1, 2 \cdots 9$

vii) If a line l is parallel to line m then m is parallel to l.

viii) $\therefore R = \{(l_1, l_1)(l_2, l_2)(l_3, l_3)(l_4, l_4)(l_5, l_5)(l_6, l_6)(l_7, l_7)$
$(l_5, l_4)(l_8, l_8)(l_9, l_9)(l_1, l_2)(l_2, l_1)(l_3, l_4)(l_4, l_3)(l_3, l_5)(l_5, l_3)(l_4, l$

ix) since $(l_i, l_i) \in R$ for $i = 1, 2, \ldots, 9$ the relation is reflexive.

x) We have $(l_1, l_2), (l_2, l_1) \in R$, $(l_3, l_4)(l_4, l_3) \in R$, $(l_3, l_5)(l_5, l_3) \in R$
$(l_4, l_5)(l_5, l_4) \in R$

$\therefore$ the relation R is symmetric.

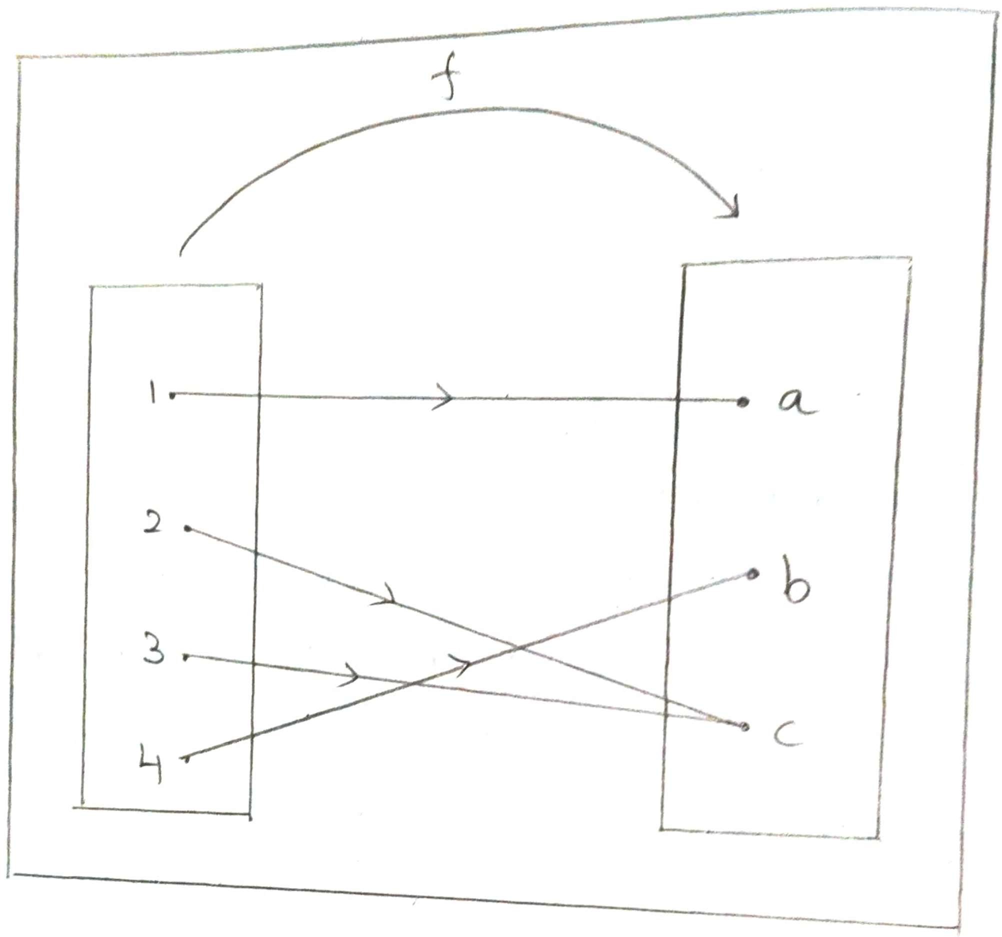
xi) We have $(l_3, l_4)(l_4, l_5) \in R$ and $(l_3, l_5) \in R$
$(l_4, l_5)(l_5, l_3) \in R$ and $(l_4, l_3) \in R$
$(l_5, l_3)(l_3, l_4) \in R$ and $(l_5, l_4) \in R$

In general if $(l_i, l_j)(l_j, l_k) \in R$, then $(l_i, l_k) \in R$

$\therefore$ the relation R is Transitive.

RESULT:

Since R is reflexive, Symmetric and Transitive, it is an equivalence relation.



f
1
2
3
4
a
b
c

ACTIVITY - 8

ONTO BUT NOT ONE - ONE FUNCTION

OBJECTIVE :

Demonstrate a function which is onto but not one - one.

PREVIOUS KNOWLEDGE:

Let A and B are non - empty sets.

i) A correspondance between the element A and B is called ~~refter~~ function from A to B if for all element of A, these corresponds a unique element of B.

ii) A function $f: A \rightarrow B$ is called a one - one function if the images of distinct elements of A are also distinct element of B.

iii) A function $f: A \rightarrow B$ is called on onto function if every element of B is the image of at least one element of A.

MATERIALS REQUIRED :

- Geometry box
- Pensils.

PROCEDURE:

i) Draw a rectangles on the paper.
ii) Draw 4 dots in the rectangles on the left sides.
iii) Draw 3 dots in the rectangle on the right sides.
iv) Mark the dots in the left rectangle as 1, 2, 3 and 4.
v) Mark the dots in the right rectangle as a, b and c
vi) Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c\}$

viii) Join elements of A to the elements of B as shown in the figure.

OBSERVATIONS:

i) In figure, to each element of A, there corresponds a unique element of B. Thus this correspondence is a function.

ii) Let 'f' represent this function.

iii) $\therefore f: A \rightarrow B$ is a function and $f(1) = a$, $f(2) = c$, $f(3) = c$, $f(4) = b$.

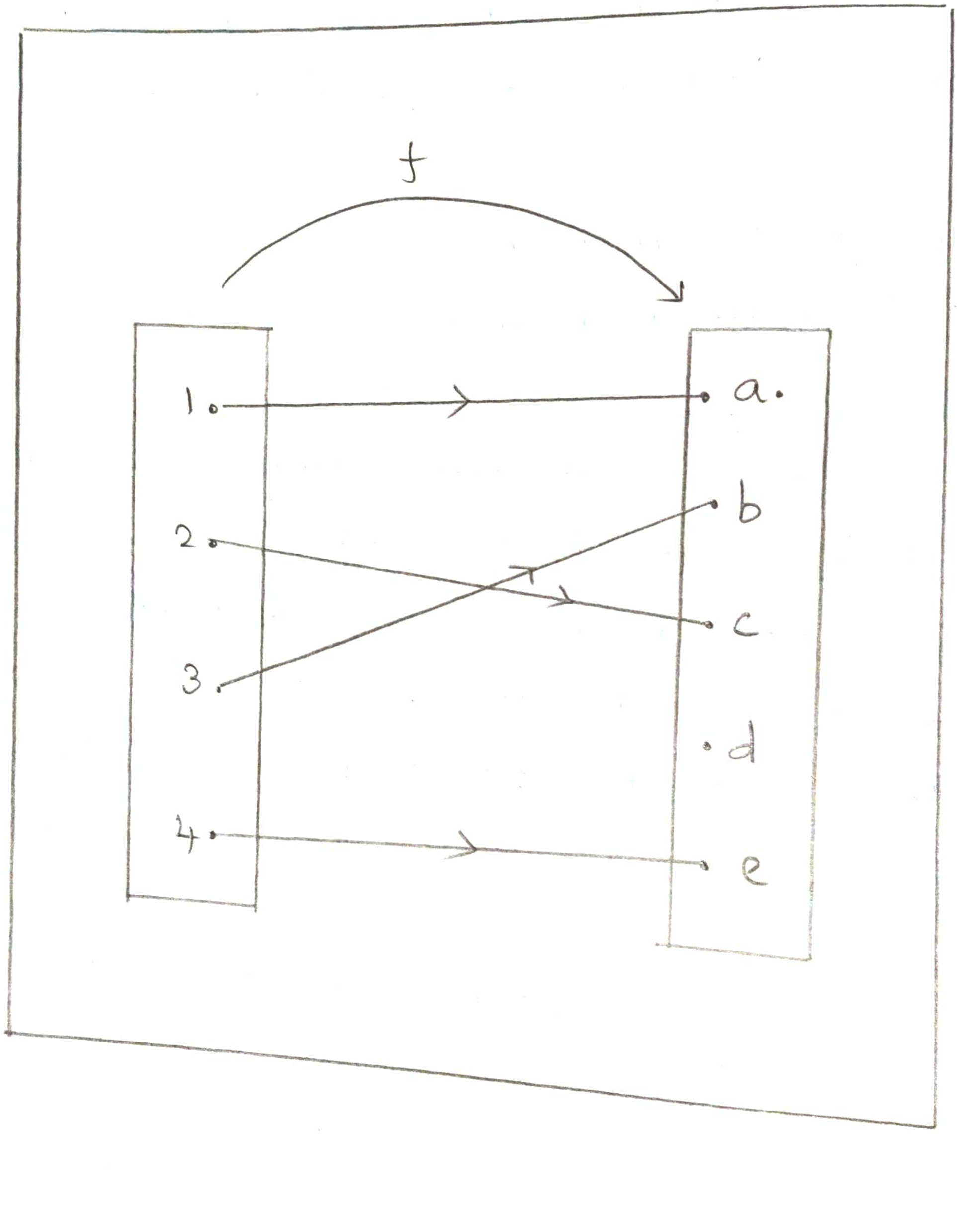
iv) Under 'f' the images of distinct elements 2 and 3 of A are not distinct because $f(2) = f(3) = c$

$\therefore f: A \rightarrow B$ is not a one-one function.

v) Every element of B is the image of at least one element of A $\therefore f: A \rightarrow B$ is an onto function.

RESULT:

$\therefore f: A \rightarrow B$ is a function which is ~~no~~ onto but not one-one.


f
1
2
3
4
a
b
c
d
e

ACTIVITY-9

ONE-ONE BUT NOT ONTO FUNCTION

OBJECTIVE:

Demonstrate a function which is one-one but not onto

PREVIOUS KNOWLEDGE:

1. Let A and B be non empty sets. A correspondence between the elements A and B is called a function from A to B if for all each element of A, there corresponds a unique element of B.
2. A function $f: A \to B$ is called a one-one function if the image of distinct elements of A are also distinct elements of B.
3. A function $f: A \to B$ is called an onto function if every element of B is the image of atleast one element of A.

MATERIALS REQUIRED:

- Geometry Box
- Pencils.

PROCEDURE:

i) Draw two rectangles on the paper.
ii) Draw 4 dots in the rectangle on the left side and mark it as 1, 2, 3 and 4.
iii) Draw 5 dots in the rectangle on the right side and mark it as a, b, c, d and e.
iv) Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c, d, e\}$
v) Join elements of A to elements of B as shown in figure.

OBSERVATION :

1. In figure, to each element of A, there corresponds a unique element of B. Thus, this correspondence is a function.
2. Let 'f' represent this function.
   $\therefore f: A \rightarrow B$ is a function and $f(1) = a$, $f(2) = c$, $f(3) = b$ and $f(4) = e$.
3. under 'f' the images of distinct elements of A are also distinct elements of B $\therefore f: A \rightarrow B$ is a one-one function
4. $d \in B$ and not any image of any element of A
   $\therefore f: A \rightarrow B$ is not an onto function.

RESULT:

$f: A \rightarrow B$ is a function which is one-one but not onto

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

ACTIVITY - 10

OBJECTIVE:

To explain the computation of conditional probability of a given event A when event B has already occured, through an example of throwing a pair of dice.

MATERIALS REQUIRED:

A4 size paper, pair of dice, glue, sketch, pen, scissors scale etc.

METHOD OF CONSTRUCTION:

1. Paste A4 sized paper.
2. Make a square and divide it to 36 units square of 1cm each
3. Write a pair of numbers as shown in figure.

DEMONSTRATION:

1. These pair of numbers gives all possible outcomes of the given experiment. Hence it represents the sample space of the experiment.
2. Here we have to find the conditional probability of an event A if an event B, has already occured, where A is the event "a number 4 appears on both dices" and B is the event "4 has appeared on at least one of the dice" i.e we have to find $p(A/B)$
3. Share outcome favourable to event A, B and $A \cap B$.

OBSERVATION:

1. Outcome favourable to A : $\{(4,4)\}$

   $n(A) = 1$

2. Outcomes favourable to B : {(1,4) (2,4) (3,4) (4,1) (4,2) (4,3) (4,4) (4,5) (4,6) (5,4) (6,4)}.

$n(B) = 11$

3. Outcomes favourable to $A \cap B$: {(4,4)}

$n(A \cap B) = 1$

4. $P(A \cap B) = 1/36$

5. $P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1}{11}$

RESULT:

To understand the concept of conditional probability, which is further used in Bayes theorem.

5/11/26